Practical: Enterprise Architecture using IDEF, ARIS and SysML/piADL

1. Title

Develop an Enterprise Architecture using IDEF, ARIS and Architecture Description Languages such as SysML/piADL.

2. Aim

To develop and represent an **Enterprise Architecture (EA)** using **IDEF, ARIS, and Architecture Description Languages (ADLs) such as SysML/piADL**, showing the relationships among business processes, applications, data, and technology infrastructure.

3. Example Problem Statement

A company wants to develop an integrated **Customer Order Management System** to manage customer orders, order processing, invoicing, databases, and business operations.
The enterprise currently has separate departments and applications for sales, finance, customer management, and inventory. The organization wants to understand how these components interact and how they can be integrated into a common enterprise architecture.
Develop an enterprise architecture that:

- Represents the major business functions and their inputs and outputs.
- Models the customer order process and organizational roles.
- Shows the relationship between applications, databases, and technology infrastructure.
- Represents information flow between different enterprise components.
- Uses **IDEF0** for functional modeling.
- Uses **ARIS** for business-process and organizational modeling.
- Uses **SysML/piADL** concepts for system and component architecture.
- Provides a layered view of **Business, Application, Data, and Technology Architecture**.

4. Software Tools Required

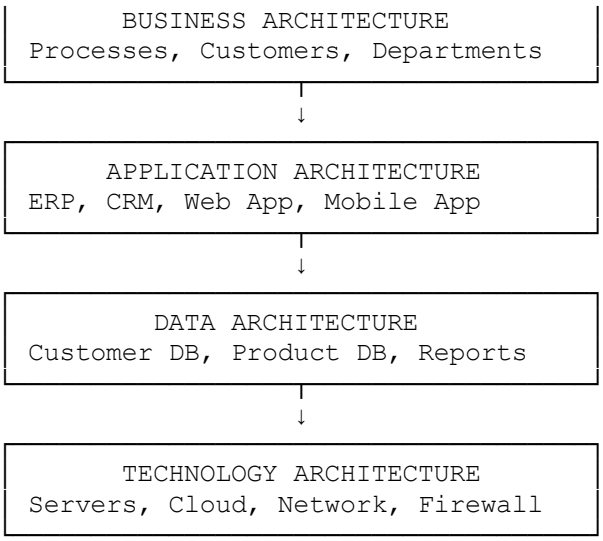
Sr. No.	Software / Tool	Purpose
1	**EdrawMax**	Creating enterprise architecture diagrams
2	Microsoft PowerPoint	Preparing and presenting architecture diagrams
3	Microsoft Word	Documentation of the practical
4	Computer/Laptop	Designing and editing the architecture

Recommended tool: EdrawMax, because it provides templates and symbols for flowcharts, UML/SysML, business process modeling, organizational charts, and architecture diagrams.

5. Theory

5.1 Enterprise Architecture

Enterprise Architecture (EA) is a structured representation of an organization that describes its **business processes, applications, data, and technology infrastructure** and the relationships between them.
The main purpose of enterprise architecture is to provide a common view of how different parts of an organization work together.
A typical enterprise architecture can be divided into four layers:



5.2 Need for Enterprise Architecture

Enterprise architecture is required to:

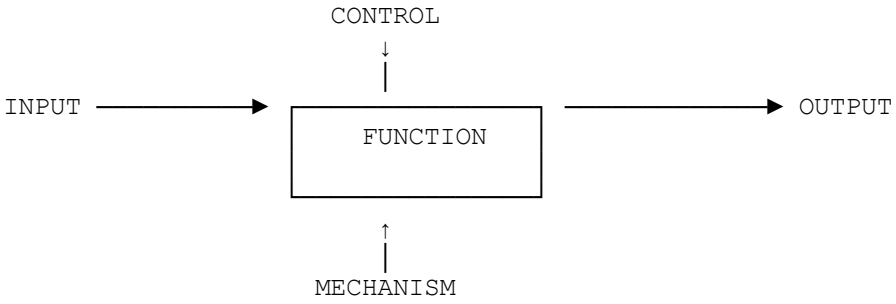
1. Understand the overall structure of an organization.
2. Identify relationships between business and IT systems.
3. Improve business processes.
4. Reduce duplication of applications and data.
5. Support integration between different systems.
6. Improve communication between business and IT teams.
7. Support strategic planning and decision-making.
8. Improve system scalability and maintainability.
9. Identify dependencies between systems.
10. Support digital transformation.

5.3 IDEF

IDEF stands for **Integration DEFinition**.
IDEF is a family of modeling methods used for representing different aspects of an organization or system.
One of the commonly used methods is **IDEF0**, which is used for **functional modeling**.
IDEF0 describes:

- What function is performed?
- What inputs are required?
- What controls govern the function?
- What outputs are produced?
- What mechanisms/resources perform the function?

IDEF0 Basic Structure



IDEF0 Components

Component	Position	Meaning
Input	Left	Data/material entering the function
Control	Top	Rules, policies, constraints
Output	Right	Result produced by the function
Mechanism	Bottom	Resources used to perform the function
Function	Center	Activity being performed

Example

For an **Order Management System**:

- **Input:** Customer order
- **Control:** Business rules and company policies
- **Function:** Process customer order
- **Output:** Confirmed order/invoice
- **Mechanism:** Sales employee, ERP system, database

5.4 ARIS

ARIS stands for **Architecture of Integrated Information Systems**.
ARIS is a framework used for modeling and analyzing **business processes and information systems**.
ARIS provides different views of an organization.
Important ARIS views include:

1. Organization View

Describes organizational units and people.
Example:

```
Customer
   ↓
Sales Department
   ↓
Finance Department
   ↓
Management
```

2. Data View

Represents information and data used by the organization.
Examples:

- Customer data
- Product data
- Order data
- Invoice data

3. Function View

Describes the functions performed by an organization.
Examples:

- Receive order

- Validate order
- Process payment
- Generate invoice
- Deliver product

4. Process View

Shows how functions, events, data, and organizational units are connected.
Example:

```
Customer Places Order
          ↓
     Process Order
          ↓
     Validate Order
          ↓
    Generate Invoice
          ↓
      Confirm Order
```

5. Output/Product View

Describes the results produced by business processes.
Examples:

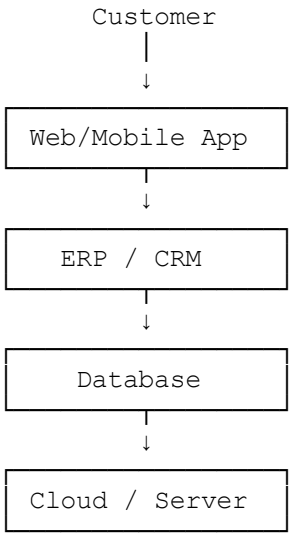
- Invoice
- Order confirmation
- Sales report
- Delivery information

5.5 SysML

SysML stands for **Systems Modeling Language**.
SysML is a modeling language used to represent the structure, behavior, requirements, and relationships of complex systems.
In enterprise architecture, SysML concepts can be used to represent:

- System components
- Subsystems
- Actors
- Interfaces
- Information flow
- Dependencies
- System boundaries

Example



Here, the customer is an **actor**, while the applications, databases, and servers represent system components.

5.6 Architecture Description Language (ADL)

An **Architecture Description Language (ADL)** is used to describe the architecture of a system in a structured manner.
An ADL generally represents:

- Components
- Connectors
- Interfaces
- Relationships
- Constraints
- System boundaries

The architecture can therefore be represented as:

```
Component A ——— Connector ——— Component B
```

For example:

```
Customer
    │
    │ Interface
    ↓
Web Application
    │
    │ API
    ↓
ERP System
    │
    │ Database Connection
    ↓
Customer Database
```

5.7 piADL

piADL can be used as an architecture description approach for describing system components and their interactions.
An architecture description focuses on identifying:

- Components
- Interfaces
- Connections
- Communication
- Data flow
- System relationships

In the proposed enterprise architecture, components such as **Customer, Web Application, ERP/CRM, Database, Analytics System, and Cloud Infrastructure** can be represented and connected.

5.8 Comparison of IDEF, ARIS and SysML/piADL

Feature	IDEF	ARIS	SysML/piADL
Main purpose	Functional modeling	Business process modeling	System architecture modeling
Main focus	Functions	Processes and organization	Components and relationships

Feature	IDEF	ARIS	SysML/piADL
Input/Output	Strongly represented	Represented through process models	Can be represented through interfaces
Organization	Limited	Strong	Can be represented
Data	Can be represented	Strong	Can be represented
Processes	Strong	Very strong	Can be modeled
Components	Limited	Moderate	Strong
Interfaces	Limited	Moderate	Strong
System structure	Limited	Moderate	Strong
Enterprise use	Functional analysis	Business process integration	System and architecture design

5.9 Proposed Enterprise Architecture

The proposed architecture for the **Customer Order Management System** contains four major layers.

Business Layer

Includes:

- Customer
- Sales Department
- Finance Department
- Order Management
- Inventory
- Management

Application Layer

Includes:

- Web Application
- Mobile Application
- ERP
- CRM
- Business Intelligence System

Data Layer

Includes:

- Customer Database
- Product Database
- Order Database
- Transaction Database
- Data Warehouse

Technology Layer

Includes:

- Cloud
- Application Server
- Database Server
- Local Network
- Firewall
- Internet

6. Methodology / Procedure

Step 1: Identify the Enterprise

Select the organization or business process to be modeled.

Example: Customer Order Management System.

Step 2: Identify Business Functions

Identify major business activities such as:

- Receive customer order
- Validate order
- Process payment
- Generate invoice
- Update inventory
- Deliver order

Step 3: Develop IDEF0 Model

Represent the main business function using:

- Input
- Control
- Function
- Output
- Mechanism

Step 4: Develop ARIS Process Model

Identify:

- Events
- Functions
- Organizational units
- Data
- Outputs

Then connect them to represent the business process.

Step 5: Develop SysML/piADL Model

Identify:

- Actors
- Components
- Interfaces
- Databases
- Servers
- Communication links

Step 6: Create Enterprise Architecture Layers

Arrange the components into:

```
Business
   ↓
Application
   ↓
Data
   ↓
Technology
```

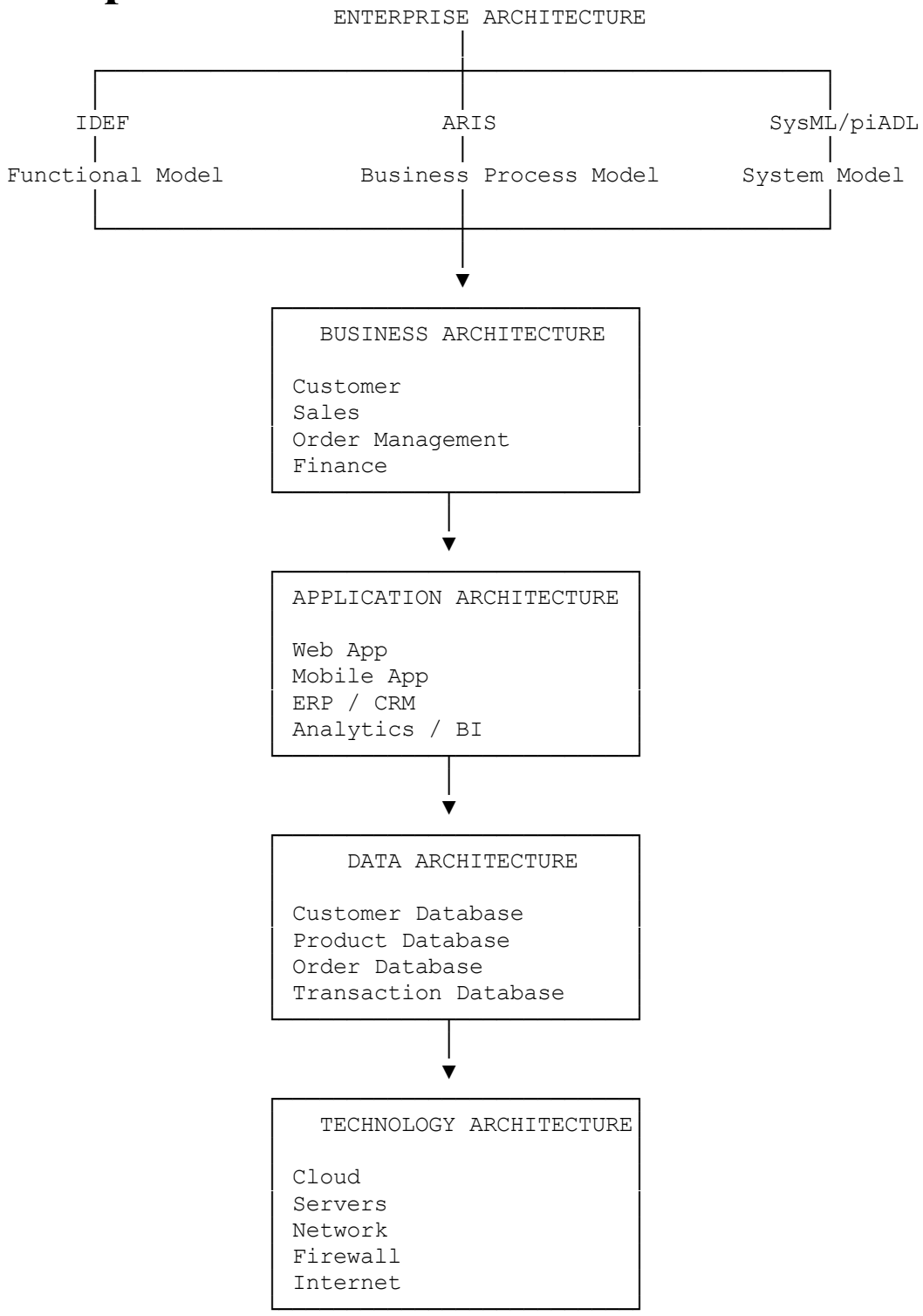
Step 7: Draw the Architecture in EdrawMax

In EdrawMax:

1. Open **EdrawMax**.
2. Select **New → Diagram**.
3. Select appropriate flowchart/UML/business-process templates.
4. Create the Business Architecture layer.
5. Create the Application Architecture layer.
6. Create the Data Architecture layer.
7. Create the Technology Architecture layer.

8. Add IDEF0 arrows.
9. Add ARIS events and functions.
10. Add SysML-style components and interfaces.
11. Connect the components using arrows/connectors.
12. Add labels and titles.
13. Check the complete architecture.
14. Export the diagram as PDF/image if required.

7. Expected Architecture



8. Result

The enterprise architecture for the **Customer Order Management System** was successfully developed using **IDEF0, ARIS, and SysML/piADL architecture description concepts**. The architecture represents the relationship between business processes, applications, data, system components, and technology infrastructure.

9. Conclusion

Enterprise Architecture provides a structured view of an organization's business and IT environment. IDEF models functions, ARIS models business processes, and SysML/piADL represents system components and relationships. Together, they provide an integrated view of the business, application, data, and technology layers.